



Good

Evening

Everyone :)

DECIMAL NUMBER SYSTEM

✓ Number we use in our day to day life.

✓ base 10 number system 0 to 9.

①

10^2	10^1	10^0
3	4	2

$$3 \times 10^2 + 4 \times 10^1 + 2 \times 10^0$$

$$300 + 40 + 2 \Rightarrow \underline{342}$$

②

10^3			
2	5	6	3

$$2 \times 10^3 + 5 \times 10^2 + 6 \times 10^1 + 3 \times 10^0$$

$$2000 + 500 + 60 + 3$$

$$\Rightarrow \underline{2563}$$

BINARY NUMBER SYSTEM

✓ The binary system, also known as base 2 system.

✓ It has only 0 and 1.

each digit represent power 2.

$$\begin{array}{ccc} 2^2 & 2^1 & 2^0 \\ 1 & 1 & 0 \end{array}$$

$$1 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 \\ 4 + 2 + 0 \Rightarrow \boxed{6}.$$

$$\begin{array}{cccc} 2^3 & 2^2 & 2^1 & 2^0 \\ 1 & 0 & 1 & 1 \end{array}$$

$$2^3 \times 1 + 2^2 \times 0 + 2^1 \times 1 + 2^0 \times 1 \\ 8 + 0 + 2 + 1 \Rightarrow 11.$$

Binary to Decimal Conversion

we need to multiply each digit of binary number by corresponding power of 2, and then add up results

eg.

1 1 0 1
 ←

$$1 \times 2^0 = 1$$

$$0 \times 2^1 = 0$$

$$1 \times 2^2 = 4$$

$$1 \times 2^3 = 8$$

$$\Rightarrow \underline{13}$$

Eg.

$$\begin{array}{cccccc} & 4 & 3 & 2 & 1 & 0 \\ 1 & 0 & \underline{1} & 0 & 1 & \end{array}$$

$$2^4 + 2^2 + 2^0$$

$$16 + 4 + 1$$

$$\Rightarrow \underline{21}.$$

Quiz:

6 5 4 3 2 1 0
1 0 1 1 0 1 0

$$0 \times 2^0 = 0$$

$$1 \times 2^1 = 2$$

$$0 \times 2^2 = 0$$

$$1 \times 2^3 = 8$$

$$1 \times 2^4 = 16$$

$$0 \times 2^5 = 0$$

$$1 \times 2^6 = 64$$

$$64 + 16 + 8 + 2$$

$$\Rightarrow \underline{90}$$

DECIMAL To BINARY CONVERSION

using long division method, we need to repeatedly divide the decimal number by 2.

and record the remainder until quotient becomes 0.

Example Convert decimal number 20 to binary

2	20	→	0
2	10	→	0
2	5	→	1
2	2	→	0
2	1	→	1
	0		

. 10100

Quiz 2.

Binary representation of 45?

2	45	→ 1
2	22	→ 0
2	11	→ 1
2	5	→ 1
2	2	→ 0
2	1	→ 1
	0	

101101

ADDITION OF DECIMAL NUMBERS

Calculate $\Rightarrow 368 + 253$

$$\begin{array}{r} 1 \quad 1 \\ 368 \\ + 253 \\ \hline 621 \end{array}$$

1) Sum = 11 . digit = $11 \% 10 = 1$, carry = $11 / 10 = 1$

2) $6 + 5 + 1 = 12$ digit = $12 \% 10 = 2$ carry = $12 / 10 = 1$

3) $3 + 2 + 1 = 6$ digit = $6 \% 10 \Rightarrow 6$ carry = $6 / 10 \Rightarrow 0$

ADDITION OF BINARY NUMBERS

①

$$\begin{array}{r}
 1 1 1 1 \\
 1 0 1 0 1 \\
 \hline
 1 1 0 1 \\
 \hline
 1 0 0 0 1 0
 \end{array}$$

① $1+1 \Rightarrow 2$ digit = $2 \% 2 = 0$ carry $\Rightarrow 2/2 = 1$

② $1+0+0 \Rightarrow 1$ digit = $1 \% 2 = 1$ carry $1/2 \Rightarrow 0$

③ $1+1 \Rightarrow 2$ digit = 0 carry = 1

④ $1+0+1 = 2$ digit = 0 carry = 1

⑤ $1+1 = 2$ digit = 0 carry = 1

Example 2

$$\begin{array}{r} \\ \\ \\ \hline 1 1 1 1 1 1 \\ \hline \end{array}$$

- | | | $(\text{Sum}/2)$ | $\text{carry} = \text{Sum}/2$ |
|---|--------------------------|--------------------|-------------------------------|
| ① | $\text{Sum} = 1$ | $\text{digit} = 1$ | $\text{carry} = 0$ |
| ② | $\text{Sum} = 1$ | $\text{digit} = 1$ | $\text{carry} = 0$ |
| ③ | $\text{Sum} = 1 + 1 = 2$ | $\text{digit} = 0$ | $\text{carry} = 1$ |
| ④ | $\text{Sum} = 0 + 0 + 1$ | $\text{digit} = 1$ | $\text{carry} = 0$ |
| ⑤ | $\text{Sum} = 1$ | $\text{digit} = 1$ | $\text{carry} = 0$ |
| ⑥ | $\text{Sum} = 2$ | $\text{digit} = 0$ | $\text{carry} = 1$ |

Quiz 2

$$\begin{array}{r} 1 1 0 \\ 1 0 1 1 0 \\ 0 0 1 1 1 \\ \hline 1 1 1 0 1 \\ \hline \end{array}$$

$$\text{Sum} = 3$$

$$d \Rightarrow 3/2 \Rightarrow 1$$

$$c \Rightarrow 3/2 \Rightarrow 1$$

BITWISE OPERATORS

Bitwise operators are used to perform operations on individual bits of binary numbers.

↳ $0 \rightarrow \text{false/unset}$ and $1 \rightarrow \text{true/set}$

AND

This operator takes two binary number and perform logical AND on their individual bits

$$\begin{array}{rcl} 6 & \rightarrow & 110 \\ 3 & \rightarrow & 011 \\ \hline & \rightarrow & 010 \end{array} \quad \begin{array}{rcl} 110 \ \& \ 011 \\ 6 \ \& \ 3 \\ \hline \rightarrow 2 \end{array} \quad \begin{array}{rcl} 6 \ \& \ 3 & \Rightarrow & 2 \end{array}$$

OR .

Takes two binary number and perform logical OR on their bits

$$\begin{array}{rcl} 0|0 & = & 0 \\ 0|1 & = & 1 \\ 1|0 & = & 1 \\ 1|1 & = & 1 \end{array} \quad \begin{array}{rcl} \text{OR } (1) & & 8|4 \end{array} \quad \begin{array}{rcl} & \Rightarrow & 1000 \\ & & 100 \\ \hline & & 1100 \end{array}$$

XOR (^)

This operator take two binary number and perform logical XOR on each pair corresponding bits

$$0 \wedge 0 \Rightarrow 0$$

$$0 \wedge 1 \Rightarrow 1$$

$$1 \wedge 0 \Rightarrow 1$$

$$1 \wedge 1 \Rightarrow 0$$

$$6 \wedge 5$$

$$2^3 \ 3 \ 2 \ 1 \ 0$$

$$\Rightarrow \begin{array}{cccc} 0 & 1 & 1 & 0 \end{array} \Rightarrow 6$$

$$\begin{array}{cccc} 0 & 1 & 0 & 1 \end{array} \Rightarrow 5$$

$$\begin{array}{cccc} 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ \hline 0 & 0 & 1 & 1 \end{array}$$

NOT (!/~)

This operator simply flip each and every bit of the number

$$\sim 0 = 1$$

$$\sim 1 = 0$$

$$\rightarrow \begin{array}{cccc} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{array}$$

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Bitwise Operations Example

5 & 6

5 $\rightarrow$ 101

6 $\rightarrow$ 110

100

= 4 .

20 | 45

20 $\rightarrow$ 010100

45 $\rightarrow$ 101101

111101

$\Rightarrow$ 61 .

92 | 154

92 $\rightarrow$ 01011100

154 $\rightarrow$ 10011010

1101110

$\Rightarrow$ 222 .

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$$\begin{array}{cccccccc} \sim & 0 & 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ \rightarrow & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 1 \end{array} \Rightarrow$$

$$2^7 \quad 2^6 \quad 2^5 \quad 2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0$$

$$128 + 32 + 2 + 1 \Rightarrow \underline{163}$$

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$$92 \wedge 154$$

$$\begin{array}{rcl} 92 & \rightarrow & 0 \cdot 1 \cdot 0 \cdot 1 \cdot 1 \cdot 1 \cdot 0 \cdot 0 \\ 154 & \rightarrow & 1 \cdot 0 \cdot 0 \cdot 1 \cdot 1 \cdot 0 \cdot 1 \cdot 0 \\ \hline & & 1 \cdot 1 \cdot 0 \cdot 0 \cdot 0 \cdot 1 \cdot 1 \cdot 0 \\ \hline & & 2 \quad 6 \quad 5 \quad 4 \quad 3 \quad 2 \quad 1 \end{array}$$

$$2^7 + 2^6 + 2^2 + 2^1$$

$$128 + 64 + 4 + 2 \Rightarrow \underline{198}$$

Quiz :

$$A \wedge B$$

$$A = 20$$

$$B = 45$$

$$\begin{array}{r} 20 \rightarrow \quad 010100 \\ 45 \rightarrow \quad \overset{1}{1}01101 \\ \hline \quad \quad 111001 \\ \hline \quad \quad 5 \quad 4 \quad 3 \quad 2 \quad 1 \quad 0 \end{array}$$

$$2^5 + 2^4 + 2^3 + 2^0$$

$$32 + 16 + 8 + 1 \Rightarrow \underline{57}$$

Binary Representation of Negative numbers.

- ↳ Convert the absolute value of number to Binary representation
- ↳ Invert all bits of the number (1's Complement)
- ↳ Add 1 to the number obtained in step 2.

$$\begin{array}{r} 5 \rightarrow \quad 0000 \ 0101 \\ \quad \quad 1111 \ 1010 \\ \quad \quad \quad +1 \\ \hline \quad \quad 1111 \ 1011 \\ \hline \end{array} \rightarrow \underline{-5}.$$

MSB $\Rightarrow$ -1 $\Rightarrow$ negative number.

$$\begin{array}{r} 0000 \ 1001 \\ 1111 \ 0110 \\ \quad \quad +1 \\ \hline 1111 \ 0111 \\ \hline \end{array} \Rightarrow \underline{-9}$$

Quiz 5.

-2

$$\begin{array}{r} 0000\ 0011 \\ 1111\ 1100 \\ +1 \end{array}$$

$$\underline{1111\ 1101} \Rightarrow -2.$$

Quiz 6. -10

$$\begin{array}{r} \rightarrow \begin{array}{cc} 0000 & 1010 \\ 1111 & 0101 \end{array} \\ \hline \phantom{\begin{array}{cc} 0000 & 1010 \\ 1111 & 0101 \end{array}} +1 \\ \hline \underline{1111\ 0110} \end{array}$$

Range of Datatype

N-bit number

$$\text{minimum} = -2^{n-1} \quad \checkmark$$

$$\text{maximum} = 2^n - 1 \quad \checkmark$$

$$32\text{-bit Integer} \quad -2^{31} \text{ to } 2^{32} - 1$$

$$2^{32} \Rightarrow 4\,294\,967\,296 \Rightarrow \underline{10 \text{ digits}}.$$

$$4 \times 10^9 \approx 10^9.$$

`int a = 105`

`int b = 106`

`int c = a * b;` 10¹¹ < `int ga = 109.`

$a = 10^5$

$b = 10^6$

`long c = a * b;`

$10^5 * 10^6 \Rightarrow 10^{11}$

X.

`int a = 105`

`int b = 106`

`long c = (long)a * b`

Importance of Constraints

$$a = 10^5$$

$$b = 10^6$$

Question $\rightarrow$ Given an array of N size,
calculate sum of all elements.

Contrainte

$$1 < N < 10^5$$

$$1 \leq A[i] \leq 10^6$$

$$1 \leq A[i] \leq 10^5$$

$$\begin{bmatrix} 10^6 & 10^6 & 10^6 & & & & & & 10^5-1 \\ & & & & & & & & 10^6 \end{bmatrix}$$

~~int~~ sum = 0
long.

```
for (int i=0; i<N; i++) {
```

$$\text{sum} = \text{sum} + A[i]$$

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point (sum)

Sum = 10¹¹ ✗ not able to hold by integer.